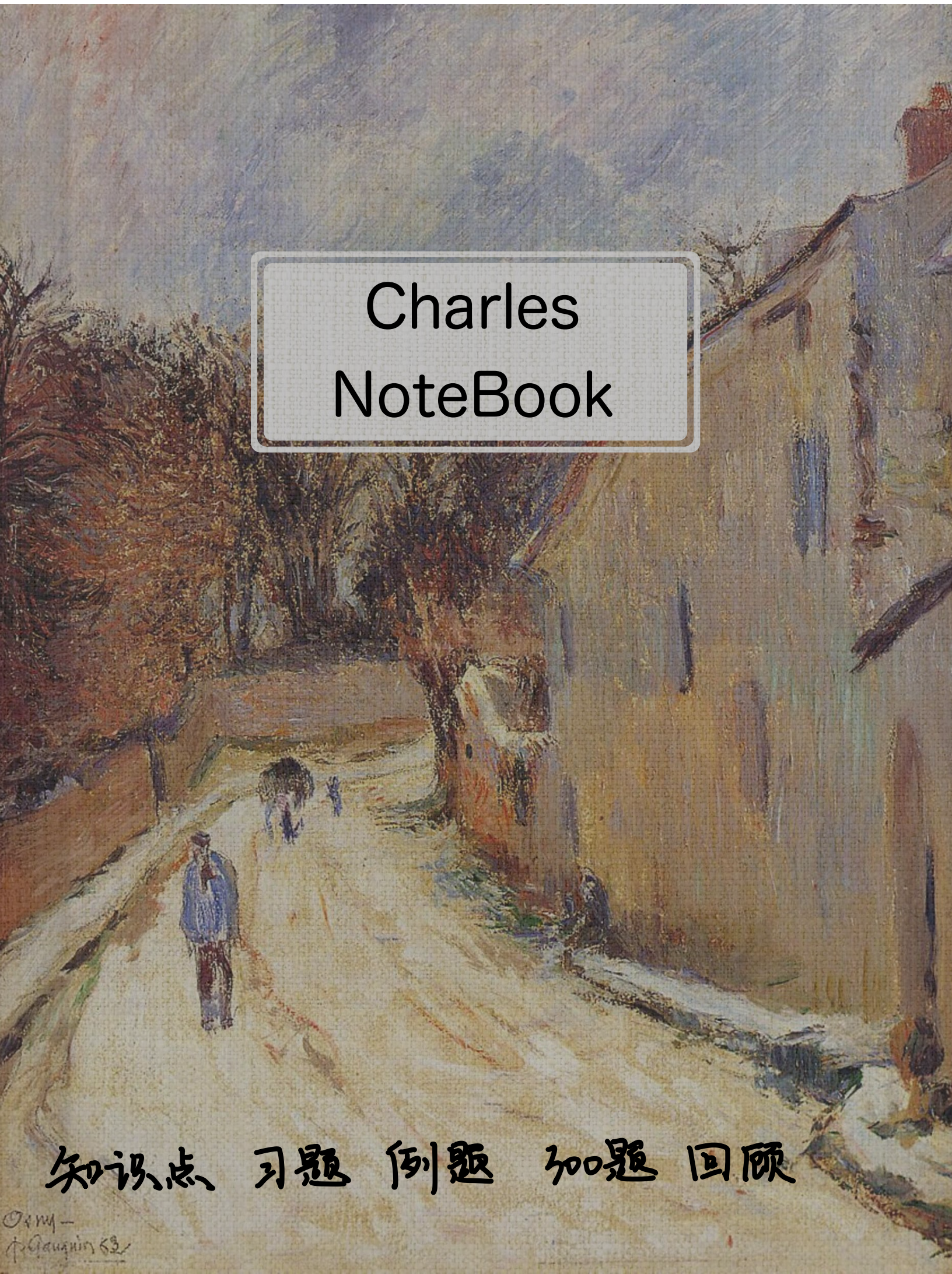




Charles NoteBook

知识点 习题 例题 加题 回顾

1. 导数定义

① 一些定义

$$\textcircled{2} \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}, \lim_{x \rightarrow x_0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

1. 导数定义

$$\textcircled{1} f'(x_0) = \lim_{x \rightarrow x_0} \frac{\Delta y}{\Delta x} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

$$\textcircled{2} \text{左导数 } f'_-(x_0) \quad \text{右导数 } f'_+(x_0)$$

$$\textcircled{3} f'(x_0) = A \Leftrightarrow f'_-(x_0) = f'_+(x_0) = A$$

$$\textcircled{4} \text{必考绝对值函数} \quad \text{eg. } y = |x| \quad \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases} \quad \begin{aligned} f'_-(0) &\triangleq \lim_{\Delta x \rightarrow 0^-} \frac{f(0 + \Delta x) - f(0)}{\Delta x} = \frac{-\Delta x}{\Delta x} = -1 \\ f'_+(0) &\triangleq \lim_{\Delta x \rightarrow 0^+} \frac{f(0 + \Delta x) - f(0)}{\Delta x} = \frac{\Delta x}{\Delta x} = 1 \end{aligned}$$

$$\textcircled{5} \text{无穷导数: } x^{-\frac{1}{2}} \quad f'(0) = \infty$$

$$\textcircled{6} \text{几何意义}$$

→ 切线; 法线

$$\textcircled{7} \text{高阶导数的意义} \quad f^{(n)}(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + n\Delta x) - f(x)}{\Delta x^n}$$

例 1.4.1

$$f(x) \text{ 在 } x=0 \text{ 可导, } f(0)=0 \quad \lim_{x \rightarrow 0} \frac{x^2 f(x) - f(x^2)}{x^2} = ?$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} - \frac{f(x^2)}{x^2} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} - \frac{f(x^2) - f(0)}{x^2 - 0} = f'(0) - f'(0) = 0$$

例 1.4.2

$$y = f(x) \text{ 在 } x=0 \text{ 可导, } f(0)=0, \lim_{x \rightarrow 0} x f\left(\frac{1}{2x+3}\right) = 1, \text{ 则 } f'(0) = ?$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{f\left(\frac{1}{2x+3}\right)}{\frac{1}{2x+3}} = \lim_{x \rightarrow 0} 2x f\left(\frac{1}{2x+3}\right) + 3 f\left(\frac{1}{2x+3}\right) = 2 + 3 f(0) = 2$$

例 1.4.3

$$f(x) \text{ 在 } x=0 \text{ 处连续} \Rightarrow f(0) = \lim_{x \rightarrow 0} f(x)$$

$$\checkmark (A) \text{ 若 } \lim_{x \rightarrow 0} \frac{f(x)}{x} \text{ 存在, 则 } f(0) = 0$$

$$\checkmark (B) \lim_{x \rightarrow 0} \frac{f(x) + f(-x)}{x} \quad \dots$$

$$\checkmark (C) \lim_{x \rightarrow 0} \frac{f(x)}{x} \quad f'(0)$$

$$\times (D) \lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{x} \quad \dots$$

$$(D) \text{ 反例 } y = f(x) = |x| \quad \lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{x} = 0 \text{ 存在, 但 } f'(0) \text{ 不存在}$$

$$f(0) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{f(x) \cdot x}{x} = A \cdot 0 = 0$$

$$A = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0)$$

$$B: \lim_{x \rightarrow 0} \frac{f(x) + f(-x)}{x} \cdot x = A \cdot 0 = \lim_{x \rightarrow 0} f(x) + f(-x) = 2f(0)$$

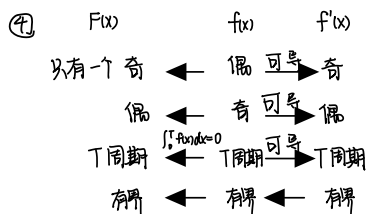
$$D: \lim_{x \rightarrow 0} \frac{f(x) - f(-x)}{x} \cdot x = A \cdot 0 = 0 \quad \lim_{x \rightarrow 0} f(x) - f(-x) = f(0) - f(0) = 0$$

得不出 $f'(0)$

$$\star \text{ 设 } f(x) \text{ 在 } x=x_0 \text{ 处连续且满足 } \lim_{x \rightarrow x_0} \frac{f(x)}{x-x_0} = A \text{ 则 } f(x_0) = 0 \quad f'(x_0) = A$$

$$\text{证明 } f(x_0) = \lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} \frac{f(x)}{x-x_0} (x-x_0) = A \cdot 0 = 0$$

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{f(x)}{x - x_0} = A$$



eg. $y = \arctan x$ $y'(0) = 0$

例 1.4.4

证明 (1) 若 $f(x)$ 是可导的偶函数则 $f'(x)$ 是奇函数

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(-x-\Delta x) - f(-x)}{\Delta x} = - \lim_{-\Delta x \rightarrow 0} \frac{f(-x-\Delta x) - f(-x)}{-\Delta x} = -f'(-x)$$

(2) 若 $f(x)$ 是可导的奇函数则 $f'(x)$ 是偶函数

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{f(-x-\Delta x) - f(-x)}{\Delta x} = \lim_{-\Delta x \rightarrow 0} \frac{f(-x-\Delta x) - f(-x)}{(-\Delta x)} = f'(-x)$$

例 1.4.5

证明 $f(x)$ 可导, 周期为 T , 则 $f'(x)$ 周期也为 T

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x+T) - f(x+T)}{\Delta x} = f'(x+T)$$

例 1.4.6

设 $f(x) = \sin x$ 可导, $\sin x$ 为周期奇函数, 且 $f(\frac{\pi}{2}) > 0$ $f'(\frac{\pi}{2}) > 0$, $M = f(\frac{\pi}{2})$

$N = f'(\frac{\pi}{2})$, $K = f''(0)$, 则 M, N, K 大小

$$K = f''(0) = 0$$

$$M = f(\frac{\pi}{2}) = 1 > 0$$

$$N = f'(\frac{\pi}{2}) = f'(\frac{\pi}{2}) = \cos(\frac{\pi}{2}) = 0$$

$$M < K < N$$

⑤ 微分

2. 微分的定义

→ 微分定义

$$\Delta y = A \Delta x + o(\Delta x)$$

线性主部 误差

$$dy = A \cdot \Delta x = f'(x) \cdot \Delta x \quad \text{又 } \Delta x = 1 \cdot \Delta x + 0 = 1 \cdot \Delta x + o(\Delta x) = \Delta x$$

$$\Rightarrow dy = f'(x) \cdot dx \Rightarrow f'(x) = \frac{dy}{dx}$$

$$dy|_{x=x_0} = f'(x_0) dx$$

→ 可微判定

① 可增量 $\Delta y = f(x_0 + \Delta x) - f(x_0)$

② 线性主部 $A\Delta x = f'(x_0) \cdot \Delta x$

③ 作极限 $\lim_{\Delta x \rightarrow 0} \frac{\Delta y - A\Delta x}{\Delta x} = 0 \Rightarrow$ 可微

例4.7

设 $y=f(x)$ 在 x_0 可导, 且 $f'(x_0) \neq 0$, 当自变量有增量 Δx 时, 函数 $y=f(x)$ 的增量为 Δy
 极限 $\lim_{\Delta x \rightarrow 0} \frac{\Delta y - dy}{dy}$ 为 ().

$$\begin{aligned} \Delta y &= dy + o(\Delta x) \\ dy &= A\Delta x = f'(x_0)\Delta x \end{aligned} \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta y - A\Delta x}{A\Delta x} = 0$$

例4.8

$f(u)$ 可导 $y=f(x^2)$, x 在 $x=-1$ 处取得增量 $\Delta x=-0.1$ 时, 相应的函数增量 Δy
 的线性主部为 0.1, 则 $f'(1) = ?$

我	$y=f(x^2) \quad g(x)=x^2 \quad y=f[g(x)]=u(x)$
的	$\Delta y = u'(-1)\Delta x^2 + o(\Delta x^2) \quad u(x) \text{ 偶}$
方	$\Delta y = \underline{2u'(-1)\Delta x} + o(\Delta x^2) \quad u(x) \downarrow \text{奇}$
法	$0.1 = 2u'(-1) \cdot (-0.1) \Rightarrow u'(-1) = -\frac{1}{2} \Rightarrow u'(1) = \frac{1}{2}$
书	$y=f(x^2)=f[g(x)], \quad y'=f'(x^2) \cdot 2x$
上	$0.1 = dy _{x=-1} = y'(-1) \cdot \Delta x$
方	$= f'(1) \cdot (-2) \cdot (-0.1)$
法	$\Rightarrow f'(1) = \frac{1}{2}$

2. 计算

① 四则运算

$$\left. \begin{aligned} (u \pm v)' &= u' \pm v' \\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2} \end{aligned} \right\} \text{及推广}$$

2. 计算

① 四则运算

$$\begin{aligned} (u \pm v)' &= u' \pm v' & d(u \pm v) &= du \pm dv \\ (u \cdot v)' &= u'v + uv' & d(u \cdot v) &= vdu + u dv \\ \left(\frac{u}{v}\right)' &= \frac{u'v - uv'}{v^2} & d\left(\frac{u}{v}\right) &= \frac{vdu - u dv}{v^2} \end{aligned}$$

① Δ 如: 设 $f(x)$ 可导, 且 $f(x)f'(x) > 0$ 则

$$(a) f(x) > f(-x) \quad (b) f(x) < f(-x) \quad (c) |f(x)| > |f(-x)| \quad (d) |f(x)| < |f(-x)|$$

$$2f(x)f'(x) = [f^2(x)]' > 0 \quad f^2(x) \uparrow \Rightarrow f^2(-x) < f^2(x) \Rightarrow |f(-x)| < |f(x)|$$

$$(u \cdot v \cdot w)' = u'vw + uv'w + uvw'$$

② Δ 如: $f'(x)g(x) - f(x)g'(x) < 0$, 其中 $g(x), f(x) > 0$, 可导函数

$$\Rightarrow \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} = \left(\frac{f(x)}{g(x)}\right)' < 0 \Rightarrow \frac{f(x)}{g(x)} \downarrow \Rightarrow f_2g_1 > f_1g_2$$

③ 例 4.9

$$\text{设 } f(x) = \sum_{n=1}^{\infty} (\tan \frac{\pi x}{4} - n) \quad \text{则 } f'(x) = ?$$

$$\text{设 } f_n(x) = \tan \frac{\pi x}{4} - n \quad \text{则 } f(x) = \sum_{n=1}^{\infty} f_n(x)$$

$$f_1(x) = \tan \frac{\pi}{4} - 1 = 0 \quad f_1'(x) = \frac{\pi}{4}$$

$$f'(x) = f_1'(x) \cdot \sum_{n=2}^{\infty} f_n(x) = -\frac{\pi}{2} \cdot 99!$$

② 分段函数

$\ln|u(x)|$: 视绝对值而不见

② 分段函数

分段点处用定义, 非分段点用公式

例 4.10

设 $y = \ln|x|, x \neq 0$ 求 y'

$$y = \begin{cases} \ln x, & x > 0 \\ \ln(-x), & x < 0 \end{cases} \quad y' = \begin{cases} \frac{1}{x}, & x > 0 \\ -\frac{1}{x}, & x < 0 \end{cases} = \frac{1}{x}, x \neq 0$$

$$\star y = \ln|x| \quad y' = \frac{1}{x} \quad \left. \begin{aligned} y &= \ln|u(x)| \\ y' &= \frac{u'(x)}{u(x)} \end{aligned} \right\} \text{视绝对值而不见}$$

例 4.12

$f(x) = 2^{1-x-a}$ 的导数

$$f(x) = \begin{cases} 2^{x-a}, & x \geq a \\ 2^{a-x}, & x < a \end{cases} \quad f'(x) = \begin{cases} 2^{x-a} \ln 2, & x > a \\ -2^{a-x} \ln 2, & x < a \end{cases}$$

$$\left. \begin{aligned} f'(a) &= \lim_{x \rightarrow a^-} \frac{2^{a-x} - 1}{x-a} = -\ln 2 \\ f'(a) &= \lim_{x \rightarrow a^+} \frac{2^{x-a} - 1}{x-a} = \ln 2 \end{aligned} \right\} \rightarrow f'(a) \text{ 不存在}$$

例 4.11

$f(x) = \begin{cases} e^{ax}, & x \leq 0 \\ b(1-x^2), & x > 0 \end{cases}$ 在 $x=0$ 处可导. $a=?$ $b=?$

$$\lim_{x \rightarrow 0^+} \frac{e^{ax} - 1}{x-0} = a$$

$$\lim_{x \rightarrow 0^-} \frac{b(1-x^2) - 1}{x-0} = 0 \quad a=0 \quad b=1$$

② 复合函数与微分形式不变性

② 复合函数与微分形式不变性

$$(f[g(x)])' = f'(g(x)) g'(x)$$

$$\frac{d f[g(x)]}{dx} = f'[g(x)] d g(x)$$

$$\Rightarrow d[f(\star)] = f'(\star) d\star$$

$$\begin{cases} (f(g(x)))' = \frac{d[f(g(x))]}{dx} \\ f'(g(x)) = \frac{d[f(g(x))]}{d g(x)} \end{cases}$$

例 4.15

设 $y = e^{\sin(\ln x)}$ 求 dy 及 $\frac{dy}{dx}$

$$dy = d e^{\sin(\ln x)} = e^{\sin(\ln x)} d(\sin(\ln x)) = e^{\sin(\ln x)} \cdot \cos(\ln x) \cdot d \ln x = \frac{e^{\sin(\ln x)} \cdot \cos(\ln x)}{x} dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{\sin(\ln x)} \cdot \cos(\ln x)}{x}$$

例 4.13

设 $y = \ln(x + \sqrt{x^2 + a^2})$, $a \neq 0$ 求 $y'|_{x=0}$

$$y' = \frac{1}{x + \sqrt{x^2 + a^2}} \quad y'|_{x=0} = \frac{1}{|a|}$$

例 4.14

$f(x) = \begin{cases} \ln x, & x \geq 1 \\ 2x-1, & x < 1 \end{cases}$ 且 $y = f(f(x))$ 求 $\left. \frac{dy}{dx} \right|_{x=e}$

$$\frac{dy}{dx} = y' = (f(f(x)))' = f'(f(x)) \cdot f'(x)$$

$$\left. \frac{dy}{dx} \right|_{x=e} = f'(f(x)) \cdot f'(x) \Big|_{x=e} = f'\left(\frac{1}{2}\right) \cdot f'(e) = 2 \cdot \frac{1}{2e} = \frac{1}{e}$$

不用写复合函数, 直接求导.

④ 反函数的导数

④ 反函数的导数

$y=f(x)$ 可导, $f'(x) \neq 0$, 则存在反函数 $x=\varphi(y)$ 且 $\frac{dx}{dy} = \frac{1}{dy/dx}$ 即 $\varphi'(y) = \frac{1}{f'(x)}$

$f'(x)$ 必恒正或恒负 (保号) $\rightarrow$ 例 6.7 奇数性质定理

$f(x)$ 单调 $\Rightarrow$ 单调函数必有反函数

$$y=f(x) \quad x=\varphi(y)=f^{-1}(y)$$

$$\varphi'(y) = \frac{dx}{dy} = \frac{1}{dy/dx} = \frac{1}{f'(x)}$$

例 4.16

$$(1) \quad y = \arcsin x \quad -1 < x < 1$$

$$x = \sin y \quad \frac{dx}{dy} = \cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \quad -1 < x < 1$$

$$(2) \quad y = \arctan x$$

$$x = \tan y \quad \frac{dx}{dy} = \sec^2 y = 1 + \tan^2 y = 1 + x^2$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$y=f(x) \Rightarrow x=\varphi(y), \text{ 记 } \begin{cases} f'(x) = y'_x \\ \varphi'(y) = x'_y \end{cases}$$

$$\text{则 } y'_x = \frac{1}{x'_y}$$

$$y''_{xx} = \frac{d^2 y}{dx^2} = \frac{d(\frac{dy}{dx})}{dx} = \frac{d(\frac{1}{x'_y})}{dx} = \frac{d(\frac{1}{x'_y})}{dy} \cdot \frac{dy}{dx} = x''_{yy} \cdot \frac{-1}{(x'_y)^2} \cdot \frac{1}{x'_y}$$

$$\Rightarrow y''_{xx} = \frac{-x''_{yy}}{(x'_y)^3}$$

eg. 满足 $\frac{dx}{dy} \neq 0$, 有微分方程 $\frac{d^2 x}{dy^2} + 3(\frac{dx}{dy})^3 = 0$ 则 $\frac{d^2 y}{dx^2} = ?$

$$\Rightarrow \frac{d^2 y}{dx^2} = \frac{\frac{d^2 x}{dy^2}}{-\left(\frac{dx}{dy}\right)^3} = 3$$

例 1.4.17

$y=f(x)$ 反函数是 $x=\varphi(y)$ 且 $f(x) = \int_1^x e^t dt + 1$, 则 $\varphi''(1) = ?$

$$y''_{xx} = \frac{-x''_{yy}}{(x'_y)^3} \quad f'(x) = 2e^{2x^2} \quad f''(x) = 2e^{4x^2} \cdot 8x = 16xe^{4x^2}$$

$$\varphi''(y) = \frac{-16xe^{4x^2}}{8e^{12x^2}} = -2xe^{-8x^2}$$

$$y=1 \text{ 时 } x=0.5 \Rightarrow \varphi''(1) = -e^{-2}$$

背 $y''_{xx} = \frac{-x''_{yy}}{(x'_y)^3}$

⑩ 变限积分求导公式

$$F(x) = \int_{\varphi_1(x)}^{\varphi_2(x)} f(t) dt \Rightarrow F'(x) = f(\varphi_2(x)) \cdot \varphi_2'(x) - f(\varphi_1(x)) \cdot \varphi_1'(x)$$

⑪ 对数求导法

$$\Rightarrow \text{多项相除, 乘, 开方, 乘方} \quad \ln y \rightarrow (\ln y)' = \frac{y'}{y}$$

例 4.20

$$\text{设 } y = \frac{\sqrt{x+2} (3-x)^4}{(x+1)^3}, \text{ 则 } y'|_{x=2} = ?$$

$$\begin{aligned} \ln y &= \frac{1}{2} \ln(x+2) + 4 \ln(3-x) - 3 \ln(x+1) \\ \frac{y'}{y} &= \frac{1}{2(x+2)} + \frac{4}{x-3} - \frac{3}{x+1} \\ y' &= \left(\frac{1}{2(x+2)} + \frac{4}{x-3} - \frac{3}{x+1} \right) \frac{\sqrt{x+2} (3-x)^4}{(x+1)^3} \\ y'|_{x=2} &= \left(\frac{1}{2 \times 4} - 4 - 1 \right) \frac{2}{3^3} = \frac{-13}{3^3} \end{aligned}$$

⑫ 幂指数函数求导法

$$u^v = e^{v \ln u} \quad (u^v)' = u^v \cdot \left(-\frac{v}{u} u' + v' \ln u \right)$$

例 4.21

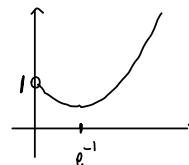
求 $y = x^x (x > 0)$ 导数

$$y = e^{x \ln x}$$

$$y' = x^x (1 + \ln x) \stackrel{=} 0 \Rightarrow x = e^{-1}$$

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{x \ln x} = 1$$

x	(0, e ⁻¹)	e ⁻¹	(e ⁻¹ , +∞)
y'	-	0	+



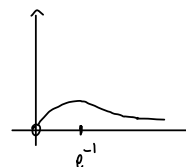
例 4.22

求 $y = x^{\frac{1}{x}} (x > 0)$ 导数

$$y = e^{\frac{\ln x}{x}} \quad y' = x^{\frac{1}{x}} \left(\frac{1 + \ln x}{x^2} \right) \stackrel{=} 0 \Rightarrow x = e^{-1}$$

$$\lim_{x \rightarrow 0^+} x^{\frac{1}{x}} = \lim_{x \rightarrow 0^+} e^{\frac{\ln x}{x}} = 0$$

$$\lim_{x \rightarrow +\infty} x^{\frac{1}{x}} = \lim_{x \rightarrow +\infty} e^{\frac{\ln x}{x}} = 1$$



如: P85 例 5.10. 求数列 $\{n^{\frac{1}{n}}\}$ 的最大项 $\Rightarrow$ 只需比较 $2^{\frac{1}{2}}$ 与 $3^{\frac{1}{3}}$ $\Rightarrow (2^{\frac{1}{2}})^6 < (3^{\frac{1}{3}})^6$
 $3^{\frac{1}{3}}$ 最大 $\Leftarrow \frac{11}{8} < \frac{11}{9}$

⑨ 高阶导数

(1) 归纳法 $eg. (2^x)^{(n)} = 2^x \cdot (\ln 2)^n$

例 4.23

$$y = \frac{1}{x} \quad y^{(n)} = (-1)^n n! x^{-(n+1)}$$

$$y = \ln x \quad y^{(n)} = (-1)^{n-1} (n-1)! x^{-n}$$

例 4.26

$$y = \frac{1-x}{1+x} \quad \text{求 } y^{(n)}(0) = ?$$

$$y' = \frac{-1-x - (1-x)}{(1+x)^2} = -2 \cdot (1+x)^{-2} \quad y = \frac{1-x}{1+x} = -1 + \frac{2}{1+x}$$

$$y'' = 2 \cdot (-1)(-2)(1+x)^{-3} \quad y^{(n)} = \dots$$

$$y^{(n)} = 2 \cdot (-1)^n n! (1+x)^{-(n+1)}$$

$$y^{(n)}|_0 = 2 \cdot (-1)^n n!$$

例 4.24

$$y = \sin x \quad y' = \cos x \quad y'' = -\sin x \quad y''' = -\cos x$$

$$\Rightarrow y^{(n)} = \sin(x + n \cdot \frac{\pi}{2})$$

$$\begin{aligned} (a^x)^{(n)} &= a^x (\ln a)^n & (\sin kx)^{(n)} &= k^n \cdot \sin(kx + n \cdot \frac{\pi}{2}) \\ (\cos kx)^{(n)} &= k^n \cos(kx + n \cdot \frac{\pi}{2}) & (\ln x)^{(n)} &= (-1)^{n-1} \frac{(n-1)!}{x^n} \quad (x > 0) \\ [\ln(x+a)]^{(n)} &= (-1)^{n-1} \frac{(n-1)!}{(x+a)^n}, & (\frac{1}{x+a})^{(n)} &= (-1)^n \frac{n!}{(x+a)^{n+1}} \end{aligned}$$

例 4.25

$$f'(x) = [f(x)]^2 \quad \text{求 } f^{(n)}(x)$$

$$f'(x) = [f(x)]^2$$

$$f''(x) = 2f(x) \cdot f'(x) = 2[f(x)]^3$$

$$f'''(x) = 2 \cdot 3 \cdot [f(x)]^4$$

$$\Rightarrow f^{(n)}(x) = n! \cdot [f(x)]^{n+1}$$

(2) 高阶求导公式

$$(uv)^{(n)} = C_n^0 u^{(n)} v + C_n^1 u^{(n-1)} v' + \dots + C_n^n u v^{(n)}$$

如 $(e^x \cos x)^{(4)} = C_4^0 e^x \cos x + C_4^1 e^x (-\sin x) + C_4^2 e^x (-\cos x) + C_4^3 e^x (\sin x) + C_4^4 e^x (\cos x)$

(2) 用泰勒公式

$$y = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n \quad x_0=0 \Rightarrow y = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

泰勒级数

抽象展开

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \dots \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \dots \\ \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots \\ \frac{1}{1+x} &= \sum_{n=0}^{\infty} (-x)^n = 1 - x + x^2 - \dots \\ \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \end{aligned}$$

具体展开

例 4.27

$$\begin{aligned} y &= x^3 \sin x \text{ 求 } y^{(6)}(0) \\ \textcircled{1} \quad y &= \sum_{n=0}^{\infty} \frac{y^{(n)}(0)}{n!} (x-x_0)^n \rightarrow x^6 \text{ 项: } \frac{y^{(6)}(0)}{6!} x^6 \\ \textcircled{2} \quad y &= x^3 \sin x = x^3 \left(x - \frac{x^3}{6} + o(x^3) \right) = x^4 - \frac{x^6}{6} + o(x^6) \\ y^{(6)}(0) &= -5! \end{aligned}$$

习 4.1

设 $f(x)$ 在 $x=0$ 处连续, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 1$ 则

$$f'(0) \stackrel{?}{=} \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{f(x)}{x} \right) \cdot x = 0$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 1 \Rightarrow \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$$

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1 \Rightarrow C$$

习 4.2

设函数 $f(x) = |x^2 - 1| \varphi(x)$, 其中 $\varphi(x)$ 在 $x=1$ 处连续, 则 $\varphi(1)=0$ 是 $f(x)$ 在 $x=1$ 处可导的?

$$f(x) = |x^2 - 1| \varphi(x)$$

$$\textcircled{1} \varphi(1) = \lim_{x \rightarrow 1} \varphi(x) = \lim_{x \rightarrow 1} \frac{f(x)}{|(x-1)(x^2+x+1)|} = \lim_{x \rightarrow 1} \frac{f(x)}{|x-1|} = 0$$

$$\left. \begin{aligned} \lim_{x \rightarrow 1} \frac{f(x)}{|x-1|} = 0 \\ f(1) = |1-1| \varphi(1) = 0 \end{aligned} \right\} \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{|x-1| - |1-1|} = 0 \Rightarrow f(1) \text{ 可导}$$

$$\textcircled{2} f(x) \text{ 可导} \Rightarrow \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x-1} \text{ 存在} \Rightarrow \lim_{x \rightarrow 1} \frac{f(x)}{x-1} \text{ 存在}$$

$$f(x) = |x^2-1| \varphi(x) \Rightarrow f(1) = 0$$

$$A = \lim_{x \rightarrow 1} \frac{|x^2-1| \varphi(x)}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1) \cdot \varphi(x)}{x-1} = \lim_{x \rightarrow 1} (x^2+x+1) \cdot \varphi(x)$$

$$\text{如果 } A \neq 0 \text{ 则 } \lim_{x \rightarrow 1} \frac{|x-1|}{x-1} \varphi(x) \neq \lim_{x \rightarrow 1} \frac{|x-1|}{x-1} \varphi(x) \Rightarrow A = 0 \Rightarrow \varphi(1) = 0$$

$\therefore$ 充分必要

习 4.3

设 $\delta > 0$, $f(x)$ 在 $[-\delta, \delta]$ 上有定义, $f(0) = 1$ 满足 $\lim_{x \rightarrow 0} \frac{\ln(1-2x) + 2xf(x)}{x^2} = 0$
证明 $f(x)$ 在 $x=0$ 可导, 求 $f'(0)$.

$$\lim_{x \rightarrow 0} \frac{\ln(1-2x) + 2xf(x)}{x^2} = \lim_{x \rightarrow 0} \frac{-2x + 2x^2 + o(x^2) + 2xf(x)}{x^2} = \lim_{x \rightarrow 0} \frac{2x(f(x) - 1)}{x^2} + 2 = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{f(x) - 1}{x} = -1$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - 1}{x} = -1 = f'(0)$$

$$\begin{aligned} \frac{1}{1+x} &\sim 1 + x + x^2 + \dots \\ \frac{1}{1-2x} &\sim 1 - (2x) + (2x)^2 - \dots \\ \ln(1-2x) &\sim 2\left[-x - \frac{x^2}{2} + \frac{4x^3}{3} + \dots\right] \end{aligned}$$

习 4.4

$y = f(\ln^2 x) e^{f(x)}$, f 可微. 计算 $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{f'(\ln^2 x) \cdot 2 \ln x}{x} e^{f(x)} + f(\ln^2 x) \cdot e^{f(x)} \cdot 2f'(x)$$

习 4.5

$f(x)$ 在 $x=2$ 某邻域内具有任意阶导数, 且 $f'(x) = e^{f(x)}$, $f(2) = 1$. 计算 $f^{(n)}(2)$ ($n \in \mathbb{Z}^+$)

$$\left. \begin{aligned} f'(x) &= e^{f(x)} \\ f''(x) &= e^{f(x)} \cdot f'(x) = e^{2f(x)} \\ f^{(n)}(x) &= e^{2f(x)} \cdot 2 \cdot f'(x) = 2e^{3f(x)} \end{aligned} \right\} \rightarrow \begin{aligned} f^{(n)}(2) &= (n-1)! e^{nf(2)} \\ f^{(n)}(2) &= n! e^{nf(2)} = (n-1)! e^n \end{aligned}$$

例 4.6

$y = y(x)$ 由 $y = x \ln y$ 确定, 求 $\frac{dy}{dx}$

$$x = \frac{y}{\ln y}$$

$$x' y = \frac{\ln y - 1}{(\ln y)^2}$$

$$y' x = \frac{(\ln y)^2}{\ln y - 1}$$

$$y' = \ln y + \frac{x}{y} y' \quad y' = \frac{y \ln y}{y - x}$$

$$y = \frac{x}{\ln x}$$

$$y' = \frac{\ln x - 1}{(\ln x)^2}, x > 0$$

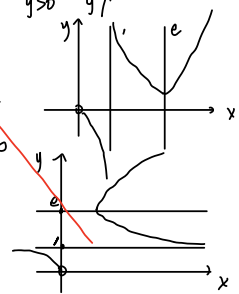
$x \in (0, 1), (0, e) \quad y' < 0 \quad y \downarrow$

$x \in (e, +\infty) \quad y' > 0 \quad y \uparrow$

$$\lim_{x \rightarrow 0} \frac{x}{\ln x} = 0$$

$$\lim_{x \rightarrow e} \frac{x}{\ln x} = e$$

$$\lim_{x \rightarrow +\infty} \frac{x}{\ln x} = +\infty$$



例 4.7

$$f(x^2) = \frac{1}{1+x^2}, \quad f'(x) = \frac{-1}{(1+x)^2}$$

$$f(x) = \frac{1}{1+x}$$

例 4.8

$y = f(\ln x) e^{f(x)}$, 其中 $f(x)$ 为可微函数, 求 dy

$$\frac{dy}{dx} = \frac{f'(\ln x)}{x} e^{f(x)} + f(\ln x) \cdot e^{f(x)} \cdot f'(x)$$

$$dy = \left[\frac{f'(\ln x)}{x} + f(\ln x) f'(x) \right] e^{f(x)} dx$$

例 4.9

$y = y(x)$ 由 $\sin(x^2 y) = xy$ 确定, $dy = ?$

$$\cos(x^2 y) \cdot (2xy + x^2 dy) = y dx + x dy$$

$$[2xy \cos(x^2 y) - y] dx = [x - x^2 \cos(x^2 y)] dy$$

$$\frac{dy}{dx} = \frac{2xy \cos(x^2 y) - y}{x - x^2 \cos(x^2 y)}$$

300.4.10

$$e^{y \ln y}$$

$y = y(x)$ 由方程 $x = y^y$ 确定, 则 dy

$$\ln x = y \ln y$$

$$\frac{dx}{x} = dy \cdot \ln y + dy \Rightarrow dy = \frac{dx}{x(1 + \ln y)}$$

300.4.11

$$f(x) = \lim_{t \rightarrow 0} x(1-2t)^{-\frac{x}{t}}, \text{ 则 } f'(x) = ?$$

$$f(x) = \lim_{t \rightarrow 0} x \left[(1-2t)^{\frac{1}{2t}} \right]^{-2x} = x e^{+2x}$$

$$f'(x) = e^{+2x} + 2x e^{+2x}$$

300.4.12

$$\text{设 } y = \sin^4 x + \cos^4 x \text{ 则 } \frac{dy}{dx} \neq 0 \text{ 时, } y^{(n)}(x) = ?$$

$$y = (\sin^2 x)^2 + 2(\sin^2 x)(\cos^2 x) + (\cos^2 x)^2 - 2(\sin^2 x)(\cos^2 x)$$

$$= (\sin^2 x + \cos^2 x)^2 - \frac{1}{2} \sin 2x$$

$$= 1 - \frac{1 - \cos 4x}{4} = \frac{3}{4} + \frac{1}{4} \cos 4x$$

$$y^{(1)} = -4^0 \sin 4x$$

$$y^{(2)} = -4^1 \sin(4x + \frac{\pi}{2})$$

$$y^{(n)} = -4^{n-1} \sin(4x + \frac{\pi}{2}(n-1))$$

300.4.13

$$\text{设 } y = e^{x^2} \text{ 求 } \frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}$$

$$\frac{dy}{dx} = e^{x^2} \cdot 2x$$

$$\frac{d^2y}{dx^2} = e^{x^2}$$

$$\frac{d^3y}{dx^3} = e^{x^2} \cdot 4x^2 + e^{x^2} \cdot 2$$

习 4.7

$$x e^{f(y)} = e^{y \ln 29}, \quad f \text{ 有二阶导数 } f' \neq 1 \quad \frac{d^2 y}{dx^2} = ?$$

$$e^{f(y)} + x e^{f(y)} \cdot f'(y) \cdot y' = e^{y \ln 29} \cdot y'$$

$$e^{f(y)} = (e^{y \ln 29} - x e^{f(y)} \cdot f'(y) \cdot y') \cdot y'$$

$$e^{f(y)} \cdot f'(y) \cdot y' = (e^{y \ln 29} - x e^{f(y)} \cdot f'(y) \cdot y') \cdot y'' + (e^{y \ln 29} - x e^{f(y)} \cdot f'(y) \cdot y') \cdot (f''(y) \cdot y') - x e^{f(y)} \cdot f''(y) \cdot y' \cdot y'$$

$$y'' = \frac{e^{f(y)} \cdot f'(y) \cdot y'}{e^{y \ln 29} - x e^{f(y)} \cdot f'(y) \cdot y' + e^{y \ln 29} - x e^{f(y)} \cdot f'(y) \cdot y' - x e^{f(y)} \cdot f''(y) \cdot y' \cdot y'}$$

两边取对数

$$\ln x e^{f(y)} = \ln e^{y \ln 29}$$

$$\ln x + f(y) = y + \ln(\ln 29)$$

$$\frac{1}{x} + f'(y) \cdot y' = y' \quad \left. \begin{aligned} -\frac{1}{x^2} + f''(y) \cdot (y')^2 + f'(y) \cdot y'' &= y'' \\ y'' &= \frac{-\frac{1}{x^2} + f''(y) \cdot (y')^2}{1 - f'(y)} \end{aligned} \right\} y'' = y''$$

习 4.8

$$f'(x) = A e^x \quad (A \text{ 为正常数}) \quad f(x) \text{ 反函数的二阶导数}$$

$$f(x) = \int f'(x) dx + C = A e^x + C = y$$

$$A e^x = \frac{dy}{dx} \quad \frac{d^2 x}{dy^2} = \frac{-\frac{dy}{dx}}{\left(\frac{dy}{dx}\right)^3} = \frac{-A e^x}{A^3 e^{3x}} = -\frac{1}{A^2 e^{2x}}$$

习 4.9

$$\text{设 } f(x) = \begin{cases} x^2 \sin \frac{\pi}{x}, & x < 0 \\ A, & x = 0 \\ ax^2 + b, & x > 0 \end{cases} \quad \text{求常数 } A, a, b \text{ 的值, 使 } f(x) \text{ 在 } x=0 \text{ 处可导, 并求 } f'(x)$$

$$\lim_{x \rightarrow 0^-} x^2 \sin \frac{\pi}{x} = 0 \quad \lim_{x \rightarrow 0^+} b = 0 \Rightarrow A = b = 0$$

$$\lim_{x \rightarrow 0^-} \frac{x^2 \sin \frac{\pi}{x} - A}{x - 0} = \lim_{x \rightarrow 0^-} x \sin \frac{\pi}{x} = 0$$

$$\lim_{x \rightarrow 0^+} 2ax = 0 \Rightarrow f'(0) = 0$$

习 4.10

设函数 $y=y(x)$ 由 $\begin{cases} x = \ln(1+t^2) + 1 \\ y = 2 \arctan t - (t+1)^2 \end{cases}$ 确定. 求 $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\frac{2}{1+t^2} - 2(t+1)}{\frac{2t}{1+t^2}} = \frac{-(t+1)(1+t^2)}{t} = -1-t-t^2$$

$$\frac{d^2y}{dx^2} = \frac{d \frac{dy}{dx} / dt}{dx/dt} = \frac{-1-2t}{\frac{2t}{1+t^2}} = \frac{(1+2t)(1+t^2)}{-2t}$$

习 4.11

设 $f(x)$ 满足 $f(0)=0$, 且 $f'(0)$ 存在, 求 $\lim_{x \rightarrow 0} \frac{f(1-\sqrt{\cos x})}{\ln(1-x \sin x)}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{f(1-\sqrt{\cos x})}{\ln(1-x \sin x)} &= \lim_{x \rightarrow 0} \frac{f(1-\sqrt{\cos x}) - f(0)}{(1-\sqrt{\cos x}) - 0} \cdot \frac{(1-\sqrt{\cos x})}{\ln(1-x \sin x)} \\ &= f'(0) \lim_{x \rightarrow 0} \frac{1-\sqrt{\cos x}}{\ln(1-x \sin x)} = f'(0) \lim_{x \rightarrow 0} \frac{1-\cos x}{-x \sin x (1+\sqrt{\cos x})} \\ &= f'(0) \lim_{x \rightarrow 0} \frac{\frac{x^2}{2} + o(x^2)}{-x^2 (1+\sqrt{\cos x})} = -\frac{f'(0)}{4} \end{aligned}$$

300.4.1

$f(x)$ 在 $x=0$ 连续, 且 $\lim_{x \rightarrow 0} \frac{f(x)}{x}$ 存在, 则?

$$\Rightarrow \left. \begin{array}{l} f(x) \text{ 与 } x \text{ 是同阶无穷小} \\ \text{极限存在} \Rightarrow \text{连续} \end{array} \right\} f(0)=0 \quad \Rightarrow C$$

300.4.2

$f(x) = x^a |x|$, a 为正整数, 则函数 $f(x)$ 在点 $x=0$ 处.

连续 $\left| \lim_{x \rightarrow 0^+} \frac{x^a |x| - 0}{x - 0} = \lim_{x \rightarrow 0^+} x^a = \lim_{x \rightarrow 0^-} -x^a = 0 \right.$ 可导 D

300.4.3

$f(x)$ 可导, $y=f(x^2)$, 当自变量 x 在 $x=-1$ 处取得增量 $\Delta x = -0.1$ 时, 相应的函数增量 Δy 的线性主部为 0.1, 则 $f'(1) = ?$

$$\Delta y = A \Delta x + o$$

$$0.1 = A(-0.1) + o \quad A = -1$$

$$\Delta y = A \cdot 2x \cdot \Delta x + o$$

$$0.1 = A(-2)(-0.1) + o$$

$$A = 0.5$$

300.4.4

$g(x)$ 在 $x=0$ 可导 $g(0) = g'(0) = 0$
 设 $f(x) = \begin{cases} \frac{g(x)}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ 则 $f(x)$ 在 $x=0$ 处

$$\textcircled{1} \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{g'(x)}{1} = 0 = f(0) \Rightarrow \text{连续}$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{g(x)}{x^2} = \lim_{x \rightarrow 0} \frac{g'(x)}{2x} = \lim_{x \rightarrow 0} \frac{g''(x)}{2} \quad \text{不一定可导}$$

当 $x \neq 0$ 时, $f'(x) = \frac{xg'(x) - g(x)}{x^2}$
 $\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} \frac{xg'(x) - g(x)}{x^2} = \lim_{x \rightarrow 0} \frac{g'(x)}{x} - \lim_{x \rightarrow 0} \frac{g(x)}{x^2}$
 $= g'(0) - \lim_{x \rightarrow 0} \frac{g(x)}{x^2} = \lim_{x \rightarrow 0} \frac{g'(x)}{x} - f'(0)$
 $\therefore f'(x)$ 在 $x=0$ 连续

300.4.5

若 $y=f(x)$ 可导, 则当 $\Delta x \rightarrow 0$ 时, $\Delta y - dy$ 为 Δx 的 A
 $\Delta y = A\Delta x + o(\Delta x)$
 $dy = A dx$

300.4.6

$f(x) = \begin{cases} x^3 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ 讨论 $f(x)$ 在 $x=0$ 处的可导性 以及 $f'(x)$ 在 $x=0$ 处的连续性

$$\lim_{x \rightarrow 0} f(x) = x^3 \sin \frac{1}{x} = 0 \Rightarrow \text{连续}$$

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

$$\lim_{x \rightarrow 0} \frac{f'(x) - f'(0)}{x - 0} = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

$$3x^2 \sin \frac{1}{x} + \frac{-x^3}{x^2} \cos \frac{1}{x}$$

可导
连续